Development and reflection

Batyrova, Yerkezhan | Matriculation no. 92131651 | DLBDSEDE02 Project: Data Engineering | Portfolio part 2

Repository: https://github.com/yerkebttt/municipal-air-quality-batch-ingestion

Implementation and user contracts

I developed a Python and MongoDB pipeline based on stakeholder actions. A dashboard receives the latest 24 records, analysts request daily pollutant trends, planners compare stations for a selected period, and data stewards inspect quality and run status. Every query is measured against a 2,000 ms target.

Reliability and data quality

The loader opens the nested UCI archive, converts each row into a timezone-aware station-hour document and writes batches of 1,000. Deterministic IDs and insert-only upserts make repeated runs safe. Invalid structures go to quarantine. Missing measurements stay absent, while doubtful values are kept with warning labels. Example thresholds only create cases for human review; they do not send an alert.

Verification and boundaries

Eleven unit tests passed. The complete Docker run stored all 420,768 rows: 383,964 were accepted and 36,804 had quality warnings, with zero structural rejection. All four benchmark queries passed the 2,000 ms target; the slowest quality summary took 221.352 ms. Docker Compose provides a reproducible environment and monitoring commands. For production, the municipality still needs verified station metadata, approved health rules, authentication, TLS, backups and interviews with actual users.